

EX NO:2b

CALCULATION AND VERIFICATION OF FREQUENCY

DEVIATION AND MODULATION INDEX OF FM SIGNALS USING MATLAB SIMULATION

Objectives:

To generate a Frequency Modulated (FM) signal in MATLAB using specified message and carrier signal parameters, and to determine the frequency deviation and modulation index of the

FM signal while analyzing the relationship between frequency deviation, message frequency,

and modulation index.

PROGRAM (MATLAB CODE)

```
% Calculation and Verification of Frequency Deviation
```

```
% and Modulation Index of FM Signal
```

```
clc;
```

```
clear;
```

```
close all;
```

```
%% Message Signal Parameters
```

```
Am = 2; % Message amplitude (V)
```

```
fm = 500; % Message frequency (Hz)
```

```
%% Carrier Signal Parameters
```

```
Ac = 1; % Carrier amplitude (V)
```

```
fc = 10000; % Carrier frequency (Hz)
```

```
%% FM Parameters
```

```
kf = 250; % Frequency sensitivity (Hz/V)
```

```
fs = 100000; % Sampling frequency (Hz)
```

```
T = 0.01; % Simulation duration (s)
```

```
%% Time Vector
```

```
t = 0:1/fs:T;
```

```
%% Message Signal
```

```

m = Am*cos(2*pi*fm*t);

%% Frequency Deviation
delta_f = kf*Am;

%% Modulation Index
beta = delta_f/fm;

%% FM Signal
fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));

%% Display Parameters
fprintf('#39;Carrier Frequency (fc) = %.0f Hz\n#39;,fc);
fprintf('#39;Message Frequency (fm) = %.0f Hz\n#39;,fm);
fprintf('#39;Message Amplitude (Am) = %.0f V\n#39;,Am);
fprintf('#39;Frequency Sensitivity (kf) = %.0f Hz/V\n#39;,kf);

fprintf('#39;\nFrequency Deviation (Delta f) = %.2f Hz\n#39;,delta_f);
fprintf('#39;Modulation Index (Beta) = %.2f\n#39;,beta);

%% Verification
beta_verify = delta_f/fm;
fprintf('#39;Verified Modulation Index = %.2f\n#39;,beta_verify);

%% Verification Check
tolerance = 1e-6;
if abs(beta - beta_verify) < tolerance
fprintf('#39;\nVerification Successful\n#39;);
else
fprintf('#39;\nVerification Failed\n#39;);
end

%% Plotting
figure;
subplot(2,1,1);

```

```

plot(t,m,'LineWidth',1.5);
grid on;
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
subplot(2,1,2);
plot(t,fm_signal,'LineWidth',1);
grid on;
title('FM Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');

```

Procedure

1. Open MATLAB and create a new script file.
2. Clear the command window, workspace variables, and close all previously opened figure

windows using:

```
Clc;
```

```
Clear;
```

```
close all;
```

3. Define the message signal parameters:

- Message amplitude $A_m=2V$

- Message frequency $f_m=500Hz$

4. Define the carrier signal parameters:

- Carrier amplitude $A_c=1V$

- Carrier frequency $f_c=10000Hz$

5. Define the frequency sensitivity constant k_f , sampling frequency f_s , and simulation

duration T

6. Generate the time vector using:

$t = 0:1/fs:T;$

7. Generate the message signal using:

$m = A_m \cos(2\pi f_m t);$

and observe the sinusoidal waveform.

8. Calculate the frequency deviation of the FM signal using:

$\Delta f = k_f \cdot A_m$

9. Calculate the modulation index using:

$\beta = \Delta f / f_m$

10. Generate the FM signal using:

$fm_signal = A_c \cos(2\pi f_c t + \beta \sin(2\pi f_m t));$

where the carrier frequency varies according to the message signal.

11. Plot the message signal in the first subplot and label the axes appropriately.

12. Plot the FM signal in the second subplot and observe the variation in instantaneous frequency.

13. Display the carrier frequency, message frequency, message amplitude, and frequency

sensitivity in the MATLAB command window using fprintf statements.

14. Display the calculated frequency deviation and modulation index values in the command

window.

15. Verify the modulation index by recomputing it using:

$$\beta = \Delta f / f_m$$

and compare the result with the previously calculated value.

16. Use a conditional statement (if-else) to check whether both modulation index values are

equal within a small tolerance.

17. Display the verification result as:

● "Verification Successful" if both values match.

● "Verification Failed" otherwise.

18. Execute the MATLAB program and observe the message signal, FM signal, calculated

frequency deviation, modulation index, and verification result.

OUTPUT:

FMSIGNALPARAMETERS

CarrierFrequency = 10000 Hz

MessageFrequency = 500 Hz

MessageAmplitude = 2.00 V

FrequencySensitivity = 1000 Hz/V

CALCULATEDVALUES

FrequencyDeviation(Δf) = 2000 Hz

ModulationIndex(β) = 4.00

Bandwidth (CarsonRule) = 5000 Hz

OUTPUT GRAPH:

